

KINESIOLOGY

Courses

KINESIOLOGY

FITNESS AND RECREATIONAL ACTIVITY

48-112 Explorations in Kinesiology

This course will introduce students to the field of kinesiology including fundamental components of health, wellness, and physical activity. Additional topics will include the history, future, and professional opportunities within the sub-disciplines: exercise physiology, biomechanics, sports nutrition, and motor behavior. Contributes to Applied Movement & Sport Studies, Exercise & Sport Studies, and Spanish for Healthcare Professions Certificate.

48-121 Structural Kinesiology

This laboratory course will provide an opportunity for students to develop an applied understanding of human anatomy and physiology with a focus on the musculoskeletal system as it relates to kinesiology, health, and wellness. Strongly encouraged to be taken concurrently with 48-123. Contributes to Applied Movement & Sport Studies, Exercise & Sport Studies, Health Studies and Neuroscience. (Fall and Spring) (NS)

48-123 Intro to Human Anatomy & Physiology

This course will provide an overview of human anatomy and physiology with a particular focus on organ systems and functions related to kinesiology, exercise, and health and wellness. Strongly encouraged to be taken concurrently with 48-121. Contributes to Applied Movement & Sport Studies, Exercise & Sport Studies, Health Studies, Neuroscience, and Spanish for Healthcare Professions Certificate. (Fall and Spring) (NS)

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team sports in educational and recreational settings. Contributes to Applied Movement & Sport Studies and Exercise & Sport Studies.

48-192 Philosophy of Coaching

This course focuses on developing general and sport-specific coaching philosophies. Topics include coaching roles, team management, relationships with constituency groups, various philosophies and theories, as well as the importance of coaching tenets. Contributes to Applied Movement & Sport Studies and Exercise & Sport Studies.

48-204 Expert Performance in Sports

Exploration of the factors that elevate athletes to the pinnacle of their disciplines. This course delves into the psychology, training, technical mastery, and professional ethics that contribute to the success of elite athletes. Through a comprehensive study of renowned sports figures and cutting-edge research, students will develop a deep understanding of the multifaceted aspects that drive exceptional athletic achievement.

48-214 Research Methods in Kinesiology

This course covers the basic concepts of research methods used in the discipline of kinesiology. It is designed to help students think critically, to give students hands-on experiences with research design, data analysis and interpretation, and to report results to a professional audience. Contributes to Data Analytics, Data Science, and Exercise and Sport Studies. Prerequisite: Mathematics 52-114 or permission of instructor. (Fall and Spring) (NS) (WA)

48-224 Care & Prevention of Athletic Injuries

This course provides an overview of allied health professions, common sport and fitness-related injuries/conditions, preventative measures, and primary means of treatment for these injuries. Topics covered include epidemiology, athletic injuries/conditions, injury prevention variables, and healthcare ethics. In addition, the course discusses essential prevention, evaluation, and care techniques for the most common injuries in the athletic setting. Hands-on skills including splinting, taping, and

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Studies and Exercise & Sport Studies. Prerequisite: Kinesiology 48-121 or permission of instructor. (Spring)

48-234 Health and Fitness Concepts

Students are presented current scientific information concerning the roles of physical activity, nutrition, and life choices in healthy living. Emphasis is placed on incorporating this information into a lifestyle that will produce lifelong optimal health. In addition, issues related to the impact of health-related diseases on society are discussed. Contributes to Applied Movement & Sport Studies, Exercise & Sport Studies, Health Studies, and Spanish for Healthcare Professions Certificate. (Fall and Spring) (NS)

48-244 Stress Management

This course is designed to provide methods and content whereby students may discover their stressors and assemble appropriate tools to help them avoid, alleviate, or rebound from stress more effectively. This class will expose the students to a holistic approach to stress management with an emphasis on the relationship between lifestyle habits and health. Students will learn both cognitive skills and relaxation techniques with the intention of preventing and/or alleviating the physical symptoms of stress. This interactive course will consist of both lecture and lab. Contributes to Exercise and Sport Studies, and Health Studies.

48-251 Integrative Approach Human A & P I

This laboratory course will provide students with an opportunity to design, execute, and analyze experiments which examine the structure and function of the human body. Experiments and activities include application and critical evaluation of human anatomy and physiology in health and disease corresponding with the topics in Human Anatomy and Physiology I. Also Biology 50-251. Contributes to Health Studies.

Prerequisites: Kinesiology 48-123/121. To be taken concurrently with 48-253. (Fall) (NS)

48-253 Human Anatomy & Physiology I

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the human body with an additional focus on integrative physiology. This first course specifically examines cells, tissue histology, and the detailed anatomy and physiology of the nervous, endocrine, integumentary, skeletal, muscular, cardiovascular, and lymphatic systems. Integrative topics include homeostasis, control systems, reflexes, body movement control, and blood pressure. Also Biology 50-253. Contributes to Health Studies. Prerequisites: Kinesiology 48-123/121. To be taken concurrently with 48-251. (Fall) (NS)

48-261 Integrative Approach Human A & P II

This laboratory course will provide students with an opportunity to design, execute, and analyze experiments which examine the structure and function of the human body. Experiments and activities include application and critical evaluation of human anatomy and physiology in health and disease corresponding with the topics in Human Anatomy and Physiology II. Also Biology 50-261. Contributes to Health Studies and Neuroscience. Prerequisites: Kinesiology 48-253/251 or Biology 50-253/251. To be taken concurrently with 48-263. (Spring) (NS)

48-263 Human Anatomy & Physiology II

This is the second of a two-course sequence in the study of human anatomy and physiology. This course is a continuation of the systematic study of the structure and function of the human body with an additional focus on integrative physiology. This second course builds upon content from Human Anatomy and Physiology I, adding to cell and tissue histology knowledge, and specifically examines the detailed anatomy and physiology of the respiratory, urinary, digestive, immune, and reproductive systems. Additional integrative topics include gas exchange and transport, fluid and electrolyte balance, metabolism, exercise, and endocrine control of growth and metabolism. Also Biology 50-263. Contributes to Health Studies and Neuroscience. Prerequisites: Kinesiology 48-253/251 or Biology 50-253/251. To be taken concurrently with 48-261. (Spring) (NS)

48-274 Fundamentals of Movement Analysis

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analysis. Students will use these skills (with emphasis placed on video analysis) to evaluate fundamental movement patterns including gait, jumping, throwing, catching and striking. Contributes to Applied Movement & Sport Studies and Exercise & Sport Studies.

48-284 Education Outreach Physical Act & Health

The purpose of this course is to build and expand on students' existing health and physical activity knowledge and enhance their abilities to communicate that knowledge with the general public. The course will focus on engaging the local community with evidence based educational outreach activities to promote health and well-being.

48-294 Comparative Physiology of Exercise

A comparative and integrative approach to athletic performance and environmental adaptations to highlight the anatomical and physiological similarities and differences between humans and other animal species. Contributes to Animal Studies. (NS)

48-302 Selected Topics

May be repeated with change in topic. Prerequisite: Permission of department chair.

48-304 Selected Topics

May be repeated with change in topic. Prerequisite: Permission of department chair.

48-314 Psychosocial Aspects Physical Activity

This course examines the psychosocial benefits of physical activity, specifically focusing on the promotion and maintenance of leisure-time physical activity. Theories of motivation, behavior change and physical activity intervention techniques as well as the importance of exercise for mental health will also be discussed. Prerequisite Kinesiology 48-234, and either Kinesiology 48-214, or Psychology 33-214 or Biology 50-232.

48-324 Motor Learning and Motor Control

This course is a study of the factors affecting the learning and control of motor skills. Emphasis is given to information processing, motor programming and motor skill

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sport studies, and neuroscience. Prerequisite: Kinesiology 48-123/121 and Kinesiology 48-214, or permission of instructor. (Spring) (NS)

48-334 Physiology of Exercise

This course focuses on musculoskeletal, nervous, cardiorespiratory, and endocrine systems as they relate to aerobic and anaerobic exercise and the metabolic response. Emphasis will be placed on the adaptations of these systems to training. Lab required. Contributes to Exercise and Sport Studies, and Neuroscience. Prerequisites: Kinesiology 48-123/121, and Kinesiology 48-214, or permission of instructor. (Fall) (NS)

48-344 Biomechanics

This course is the study of basic anatomical and mechanical principles applied to human movement. Emphasis is placed on kinematic and kinetic concepts and the use of computerized movement analysis. Lab required. Contributes to Exercise and Sport Studies. Prerequisites: Kinesiology 48-123/121 and Kinesiology 48-214, or permission of instructor. (Spring) (NS)

48-374 Nutritional Sciences

This course examines the physiological and cellular functions of energy macronutrients and micronutrients. Topics include metabolism and energy balance, life cycle nutrition, nutrition for athletic performance, and the function of nutrients as they relate to human health and disease. Prerequisite: Kinesiology 48-234, or permission of instructor. Contributes to Applied Movement & Sport Studies, Exercise & Sport Studies, and Health Studies. (NS)

48-394 Seminar in Kinesiology

This course is designed to investigate special topics or problems of interest in kinesiology. Areas of study will be selected by both instructor and students with students taking an active role as both learners and teachers. Prerequisites: Kinesiology 48-214, and at least one of the following: Kinesiology 48-324, 48-334, or 48-344, or permission of instructor.

[Menu](#) [Search](#)[Apply](#) [Visit](#) [Majors & Minors](#)

The purpose of this course is to provide students with the skills needed to develop, implement and evaluate health promotion programs with a special emphasis on behavior change initiatives. Prerequisite: Kinesiology 48-134 and 48-214.

48-434 Theory & Practice Strength & Condition

An overview of strength and conditioning principles. Emphasis is placed on the exercise sciences (including anatomy, exercise physiology, and biomechanics) and nutrition, exercise technique, program design, organization and administration, and testing and evaluation. The course prepares students for the nationally accredited Certified Strength and Conditioning Specialist (CSCS) certification exam. Contributes to Applied Movement & Sport Studies. Prerequisites: Kinesiology 48-123/121 or 48-253/251.

48-444 Exercise Prescription

This course is the study of current practices related to fitness assessment and exercise prescription. Emphasis is placed on the collection of data from fitness testing and the design of personalized exercise programs. Contributes to Exercise and Sport Studies. Prerequisites: Kinesiology 48-214 and 48-334. (Spring) (NS)

48-454 Tissue Mechanics

This is an advanced course that explores the anatomical structure and physiological and mechanical function of tissues within the body including bone, cartilage, tendons, ligaments, nerves and muscle. Special attention is given to mechanical testing of these materials with application of this information to the study of injury mechanisms and diseases of these tissues. Also Biology 50-454. Contributes to Health Studies.

Prerequisites: Kinesiology 48-123/121, and either Kinesiology 48-344 or Physics 53-154, or permission of instructor. (NS)

48-474 Muscle Physiology

This course is designed to enhance the student's knowledge of the field of muscle physiology. The course focuses on the functional properties of skeletal muscle as it impacts health and performance in normal and pathological situations with specific

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48-481 Introduction to Capstone

This is the first of two courses in the research capstone sequence. In this course a student will develop a laboratory or field-based research experience that includes hands-on experience directed toward answering questions related to kinesiology. In this writing-intensive course, students work directly with a faculty member to develop a research proposal and obtain university approval for human participant research, if necessary. Prerequisites: Kinesiology 48-214, and either 48-334 or 48-344. (Spring) (WA)

48-483 Experiential Capstone in Kinesiology

This is the second of two courses in the capstone sequence in which the student will complete a field-based research project as proposed in 48-481 that is designed to provide the student a hands-on experience directed toward answering a question related to kinesiology. Each student will formally present results of the study in oral and written form. Prerequisites: Kinesiology 48-481. (Fall) (WA)

48-493 Experimental Capstone in Kinesiology

This is the second of two courses in the capstone sequence in which the student will complete a laboratory research project designed to provide hands-on experience directed toward answering a question related to kinesiology. In this course, students will collect and analyze experimental data for their capstone projects as proposed in 48-481. Each student will formally present results of the study in oral and written form. Prerequisites: Kinesiology 48-481. (Fall) (WA)

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